

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Senior Software Engineer, Perception

Senior Perception Engineer with over three years of experience building end-to-end CNN perception systems for real-time, safety-critical applications. Experienced with object detection, LiDAR odometry, and deterministic real-time inference on edge and cloud platforms. Skilled in Python, C++, and ROS2, with a background in leading teams and deploying AI systems on the edge. I recently completed an M.Sc. in Robotics at the University of Bristol where I deepened my knowledge of CNN models, blending solid research skills with a hands-on engineering approach.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Built and deployed end-to-end CNN models using YOLOv5 to enhance worker and key equipment detection accuracy by 15% in industrial safety systems.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Built an automated sensor-data cleansing pipeline processing 3.6 million data points, reducing model rollout time from 3 days to under 24 hours and improving data quality for downstream CNN training.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of CNN and time-series models.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Built and optimised deep learning pipelines with YOLOv5 and DeepSORT, enabling consistent object tracking and robust analytics in dynamic industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing throughput by 233% and enabling real-time detection of safety-critical events.
- Developed synthetic data generation and augmentation tools using OpenCV and NumPy, accelerating model iteration cycles and improving edge-case coverage.
- Implemented rigorous testing and validation frameworks simulating challenging lighting, occlusion, and motion conditions, improving CNN stability and reducing variance by 25%.
- Deployed a unified quantised CNN model for IR and RGB sensor streams using Intel OpenVINO, ensuring deterministic real-time inference on constrained edge devices.
- Collaborated closely with software, hardware, and QA teams to integrate AI modules into production systems, delivering scalable, fault-tolerant solutions for engineering safety applications.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and RGB data for autonomous navigation on an Unmanned Ground Vehicle (UGV).
- Processed and analysed large-scale point cloud datasets using Open3D and RTAB-Map to perform 3D feature extraction, registration, and map optimization.
- Tuned SLAM parameters to improve localization accuracy across varied terrains, achieving stable performance in both feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS 2 before successful real-world deployment.
- Integrated multiple sensing modalities to achieve robust spatial and radiological understanding, enabling accurate obstacle detection in static environments.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Core Machine Learning and Optimisation:** Convolutional Neural Networks (YOLOv5/v8, EfficientDet), DeepSORT, OpenVINO, TensorRT, ONNX, Optuna, MultiRocket, Bayesian Optimisation
- **Perception & Computer Vision:** OpenCV, LiDAR SLAM, Open3D, Point Cloud Library, 3D Feature Extraction, Registration, Denoising, NumPy, Scikit-Image, Data Augmentation
- **Cloud, Edge & MLOps:** AWS, Microsoft Azure, MLFlow, Intel VPUs, NVIDIA Jetson, FastAPI, Apache Kafka, DVC (Data Version Control), Docker, Linux, Git
- **Programming & Development:** Python, C++, Bash, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), Gazebo (Harmonic), Nav2, Pandas, PostgreSQL